A

Project Work-2 Report

on

IOT BASED RAILWAY TRACK DEFECT DETECTION

Submitted in Partial Fulfillment of the Academic Requirements of Degree

Bachelor of Engineering

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

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MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE

(Autonomous**)**

(Accredited by NBA & NAAC)

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April, 2025

DECLARATION

This is to certify that the Project work-2 Report titled, **“IoT based Railway track defect detection ”,** being submitted by, **Y. Lohitha (2451-21-735-135), P. Sabitha (2451-21-735-171), P. Usha (2451-21-735-316),** in partial fulfillment for the award of Bachelor of Engineering (BE) degree, with specialization of Electronics and Communication Engineering, to the Department of Electronics and Communication Engineering, *MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE, an autonomous institution under OSMANIA UNIVERSITY*, *Hyderabad,* is a record of the original work carried out by me under the supervision of **K. Aruna.** Further this is to state that the results embodied in this project report have not been submitted to any other University or institution for the award of any Degree or Diploma.

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CERTIFICATE

This is to certify that the Project work-2 Report titled, **"IoT Based Railway Track Defect Detection",** being submitted by, **Y. Lohitha(2451-21-735-135), P. Sabitha(2451-21-735-171), P. Usha(2451-21-735-316),** in partial fulfillment for the award of Bachelor of Engineering (BE) degree, with specialization Electronics and Communication Engineering (ECE), to the Department of Electronics and Communication Engineering, *MATURI VENKATA SUBBA RAO (MVSR) ENGINEERING COLLEGE, an autonomous institution under OSMANIA UNIVERSITY*, *Hyderabad,* is a record of the bona fide work carried out by him/her under my guidance and supervision.

Signature of the Guide	**Head of the Department**
K. Aruna	Dr. G. Kanaka Durga,
Sr. Assistant Professor	Professor & HECE

External Examiner

ACKNOWLEDGEMENTS

I sincerely express my heartfelt gratitude to my supervisor, **K. Aruna, Sr. Assistant Professor** for their invaluable guidance, unwavering support, and insightful contributions throughout this Project work- 2. The discussions we had greatly enhanced our understanding and played a crucial role in achieving our project goals.

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Furthermore, I would like to acknowledge the faculty and staffs of the department for their direct and indirect assistance in making this project a success.

Lastly, I am deeply grateful to my parents for their unwavering support, encouragement, and blessings throughout this journey.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	DESCRIPTION
1.	LCD	Liquid Crystal Display
2.	Esp32	Espressif
3.	Node MCU	Node micro-controller unit
4.	GPS	Global Positioning System
5.	GSM module	Global System for Mobile Communication

ABSTRACT

The rapid growth of railway transportation systems has necessitated the development of advanced safety mechanisms to prevent accidents caused by track defects. One of the leading causes of train derailments and accidents is the presence of cracks, misalignments, or other structural issues on railway tracks. Traditional track inspection methods are often manual, time-consuming, and prone to human error. To address these limitations, this paper proposes an IoT-based Railway Track Defect Detection System that leverages smart sensors and wireless communication technologies for real-time monitoring and reporting of track anomalies. The proposed system consists of a mobile unit mounted on a railway inspection vehicle or a small robot that traverses the railway tracks. It is equipped with ultrasonic sensors, ESP32 camera, to detect various forms of defects such as cracks, displacements, and surface wear. These sensors collect real-time data as the unit moves along the track. A microcontroller, such as an ESP32 and node MCU can be used.

Once a defect is detected, the system uses IoT modules like GSM, GPS, or LoRa to send the information to a centralized cloud platform or monitoring station. The transmitted data includes the type of image defect and its exact geographical location, allowing maintenance teams to take prompt and targeted action. The integration of GPS ensures that the location of defects is accurately mapped and stored for analysis. The system offers several advantages over conventional methods: it reduces inspection time, increases accuracy, and provides real-time alerts, thereby enhancing overall safety and maintenance efficiency. Furthermore, the data collected can be used for predictive maintenance and long-term infrastructure planning. This IoT-based solution not only ensures safer railway operations but also contributes to the modernization of railway infrastructure with smart technologies.

CHAPTER -I

INTRODUCTION

1.1 Introduction

Railways play a vital role in the transportation infrastructure of any country, ensuring the efficient movement of goods and people. However, the safety of railway networks is often compromised by cracks in the tracks and unexpected obstacles, which can lead to severe accidents, loss of life, and financial damages. Regular inspection and maintenance of railway tracks are therefore essential, yet traditional methods are time-consuming, labor-intensive, and may not always detect issues promptly the safety and reliability of railway transportation are crucial for the economic and social development of a country. With the expansion of rail networks and increasing demand for high-speed travel, ensuring the integrity of railway infrastructure has become more critical than ever.

Among the various causes of railway accidents, track-related faults such as cracks and the presence of foreign obstacles are among the most common and dangerous. These issues, if undetected, can lead to derailments, service disruptions, and severe loss of life and property. To address these challenges, this project presents the design and development of a Railway Track Crack and Obstacle Detection Robot. The system is a moving robotic unit that autonomously travels along the railway track, continuously scanning for cracks and obstacles. The robot is equipped with an ultrasonic sensor to detect surface cracks on the tracks and identify the presence of any foreign objects. Additionally, an ESP32-CAM module captures real-time images of the detected faults, aiding in visual verification. When a crack or obstacle is detected, the robot instantly sends the exact geographical location via SMS using the GSM module, while the GPS module ensures accurate position tracking. This enables maintenance teams to act quickly and precisely, reducing response time and ensuring safer railway operations. This project demonstrates a cost-effective, autonomous solution to improve railway safety and operational efficiency through real-time monitoring, location tracking, and image-based evidence.

1.2 Objectives of the project

The primary objective of this major project is to design, develop, and implement an autonomous robotic system capable of efficiently and reliably detecting critical anomalies on railway tracks, specifically cracks and obstructions. This system integrates a NodeMCU microcontroller for real-time data processing and control, ultrasonic sensors for non-contact crack and obstacle detection, a GPS module for precise location identification of detected anomalies, and a GSM module for immediate remote reporting. Furthermore, the inclusion of an ESP32-CAM module enables visual confirmation of detected issues through captured images transmitted alongside location data. The successful realization of this project aims to contribute to enhanced railway safety by providing a proactive and automated solution for track inspection and timely identification of potential hazards, thereby minimizing the risk of accidents and ensuring smoother, safer railway operations.

Building upon the foundation of enhanced railway safety and proactive anomaly detection, the extended objective of this project encompasses the seamless integration of the autonomous robotic system into existing railway infrastructure and operational workflows. This includes:

- Developing a robust and user-friendly data visualization and analysis platform: This platform will receive real-time reports of detected cracks and obstructions, along with their precise GPS coordinates and visual confirmation, enabling railway authorities to efficiently monitor track conditions, prioritize maintenance schedules, and analyze trends in anomaly occurrences over time.
- Ensuring system scalability and adaptability to diverse railway environments: The project aims to create a system that can be easily deployed across various track types, terrains, and operational conditions, accommodating potential variations in track geometry and environmental factors.
- Implementing power-efficient design and sustainable operation: The extended objective includes optimizing the system's power consumption for extended operational periods and exploring sustainable power sources to minimize its environmental impact and reduce maintenance requirements.

- Conducting rigorous field testing and validation in real-world railway scenarios: Comprehensive testing under diverse operational conditions will be crucial to evaluate the system's reliability, accuracy, and robustness, ensuring its effectiveness in practical railway environments.
- Exploring the potential for predictive maintenance capabilities: By analyzing historical data of detected anomalies and correlating them with factors like track usage, environmental conditions, and maintenance history, the project aims to investigate the feasibility of developing predictive maintenance algorithms to anticipate potential track failures before they occur.
- Investigating the integration of advanced sensor technologies: Future extensions of this project could explore incorporating more sophisticated sensors, such as vibration sensors or thermal imaging, to provide a more comprehensive understanding of track conditions and detect a wider range of potential issues.

1.3 Problem Statement:

The current manual methods of railway track inspection are often time-consuming, labor-intensive, and prone to human error, leading to potential delays in identifying critical anomalies such as cracks and obstructions. These undetected issues can pose significant safety risks, potentially causing accidents and disruptions to railway operations. Furthermore, the lack of real-time location data and visual confirmation of detected anomalies can hinder swift and effective maintenance responses. This project addresses the critical need for an automated, reliable, and efficient system for early detection and reporting of railway track cracks and obstructions, incorporating precise location identification and visual verification to enhance railway safety and operational efficiency.

1.4 Scope of the Thesis:

This thesis explores the development of a small, mobile robot designed to automatically detect cracks and obstacles on railway tracks using readily available components. The robot integrates a NodeMCU for central control, an ESP32-CAM for visual inspection, a UV sensor to potentially identify surface anomalies related to cracks, a GPS module to pinpoint the location of detected issues, and a GSM module to send SMS alerts containing the problem type and its coordinates.

The project will focus on designing the robot platform, implementing algorithms to identify cracks and obstacles from visual and UV sensor data, managing location tracking, and establishing wireless communication for alerts. The system's performance will be evaluated in a controlled setting, aiming to demonstrate the feasibility of a low-cost, multi-sensor approach for enhancing railway safety through automated inspection and timely reporting of potential hazards.

1.5 Organization of the Thesis:

Chapter 1: Introduction

This chapter introduces the critical issue of railway track safety and the potential dangers posed by undetected cracks and obstructions. The chapter clearly states the primary objective of this project: to design and develop an autonomous system for real-time detection of cracks and obstacles on railway tracks, utilizing GPS for precise location identification, GSM for immediate remote reporting, and an ESP32-CAM for visual confirmation.

Chapter 2: Literature Survey

This chapter provides a comprehensive review of existing research and technologies related to railway track inspection and anomaly detection. It will specifically justify the selection of ultrasonic sensors for crack and obstacle detection, GPS for localization, GSM for communication, and the ESP32-CAM for visual verification, highlighting their suitability and advantages for this project's objectives compared to alternative solutions.

Chapter 3: Theoretical aspects of the project

This chapter details the systematic approach employed in the development of the autonomous railway track inspection system. It outlines the functional architecture of the system, explaining the integration of the NodeMCU microcontroller, ultrasonic sensors, GPS module, GSM module, and ESP32-CAM. The capture and transmission of visual information using the ESP32-CAM. It also describes the data flow and communication protocols between the different modules.

Chapter 4: Results

This chapter presents and analyzes the results obtained from testing the developed system. It will evaluate the system's performance in terms of its accuracy in detecting cracks and obstacles, the reliability of location identification using GPS, the effectiveness of remote reporting via GSM, and the utility of the visual information provided by the ESP32-CAM. The chapter will discuss the strengths and limitations of the system based on the experimental findings. It may also compare the performance of this system with existing methods or similar research, highlighting its advantages and potential areas for improvement.

Chapter 5: Summary & Conclusion

This chapter lays the groundwork for a system designed to automatically detect cracks and obstructions on railway tracks. It leverages the ESP32-CAM for image acquisition, Node MCU for overall system control and data processing, GPS for location tagging of detected anomalies, and GSM for remote communication of alerts.

CHAPTER-II

LITERATURE SURVEY

2.1 Literature Review

IOT BASED VISUAL DEFECT DETECTION IN RAILWAY TRACKS

Authors: A. Geethanjali , T. Vijitha , S. Naveen Kumar , D. Gajendran , U. Thulasi Ram , N. Moulika

Abstract: This paper proposes the design of crack finding robot in the railway tracks. In India rail transportation engage a major pose in asset with the essential transportation to maintain necessities of a briskly emergent financial system. At present, India possesses the 4th major railway network in the globe.It has proven inappropriate for the control system currently use in Indian railways. Therefore, there is a need to have new technology which will be vigorous, well organised and stable for both crack detection in railway track as well as object or things detection. A robust monitoring system has been suggested and clarified in this paper to address the shortcomings of the existing rail surveillance system to detect cracks of the railway tracks. To detect cracks & damages in railway tracks. To design & develop IoT enabled Robo. To inform Railway Controller about cracks & damaged track information when detected by the robot. To send all the information through wireless communication based devices. Designing the project to overcome this problem.

RAILWAY TRACK CRACK DETECTION

Authors: Arun Kumar R,Vanishree K, Shweta K,Nandini C, Shweta G

Abstract: In India railways transportation service is the cheap and the majority convenient mode of passenger transport and also for long distance and suburban traffic. The main cause of the accidents happened in railways are railway track crossing and unrevealed crack in railway tracks. Therefore, there is a need to have new technology which will be robust, efficient and stable for both crack detection in railway track as well as object detection. This project discusses a Railway track crack detection using sensors and is a dynamic approach which combines the use of GPS tracking system to send alert messages and the geographical coordinate of location. Arduino Microcontrollers used to control and coordinate the activities of this device.

2.1.1 Existing Methods:

Existing Methods for Railway Track Defect Detection – In Detail Railway tracks are critical infrastructure, and their regular inspection is vital to avoid serious accidents. Over the years, several technologies and manual techniques have been employed to detect defects like cracks, misalignments, and breaks in railway tracks. Below are the main existing methods, their working principles, and drawbacks:

2.1.1.1 Manual Track Inspection

Working Principle: Track inspectors walk along the railway lines to look for visible defects. Sometimes, they use simple tools like hammers to listen for changes in sound that indicate cracks.

Advantages:

- Low cost
- Does not require advanced technology

Drawbacks:

- Time-consuming and labor- intensive
- Inefficient for long distances
- Prone to human error and fatigue
- Not suitable for real-time or frequent inspection
- Dangerous in remote or active railway zones

2.1.1.2 Ultrasonic Rail Testing (URT)

Working Principle:

- Ultrasonic sensors send sound waves into the rail. If there is a crack or internal flaw, the wave will reflect back differently, allowing for detection.

Advantages:

- Can detect internal (non-visible) flaws
- More accurate than manual inspection

Drawbacks:

- Expensive and bulky equipment
- Requires specialized operators
- Only available for scheduled inspections

2.1.1.3 Laser-Based and Magnetic Induction Testing

Working Principle:

- Laser scanners or magnetic sensors detect surface irregularities and misalignments.

Advantages:

- Non-contact and high-speed inspection
- Can be mounted on inspection vehicles

Drawbacks:

- Very high cost
- Complex maintenance
- Not feasible for developing countries or rural areas

2.1.1.4 Drone-Based Visual Inspection

- Working Principle:
- Drones equipped with cameras capture aerial images or videos of railway tracks for offline or AI-based analysis.

Advantages:

- Covers large areas quickly
- No physical contact with the track

Drawbacks:

- Limited battery life
- Weather-dependent
- Expensive and requires skilled pilots
- Not suitable for continuous, track-level monitoring

2.1.2 Proposed System:

The proposed system is an IoT-enabled robotic inspection unit designed to autonomously detect cracks or defects in railway tracks using ultrasonic sensing technology. It uses an ESP32 microcontroller as the brain of the system, which interfaces with ultrasonic sensors to detect physical anomalies on the track surface. When a defect is detected, the system captures the location using a GPS module and sends an alert via GSM to a monitoring station or mobile phone. This system provides a real-time, low-cost, and efficient solution for railway safety monitoring.

2.2 Summary:

The literature reveals an evolution from manual inspection and basic sensor-based methods for railway track and crack detection towards more sophisticated automated systems. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), have demonstrated significant promise in accurately identifying and localizing track defects due to their capacity to learn complex visual patterns from track imagery. Integrating CNN-based feature extraction with robust classifiers like Support Vector Machines (SVM) or employing end-to-end CNN architectures presents a compelling approach to address the challenges of reliable and efficient railway infrastructure monitoring.

The proposed methodology, which leverages [mention specific techniques or architectures you are focusing on, e.g., advanced CNN architectures, sensor fusion techniques, etc.] for defect detection and classification, is supported by a strong foundation of prior research and empirical findings. This approach offers the potential for a scalable, accurate, and timely solution for railway maintenance, contributing to enhanced safety and operational efficiency. By building upon existing research and addressing limitations in current methodologies, this project aims to advance the field of automated railway track inspection for improved infrastructure integrity. The subsequent sections will detail the specific methodologies employed in the design and implementation of this detection system.

CHAPTER-III

THEORETICAL ASPECTS OF THE RAILWAY DEFECT DETECTION SYSTEM

3.1 Theoretical concepts

This project leverages the principles of robotics, sensor fusion, and the Internet of Things (IoT) to create an autonomous system for enhancing railway safety and maintenance efficiency. The core concept revolves around deploying a mobile robot equipped with a suite of sensors to traverse railway tracks, identify critical anomalies (cracks and obstacles), and communicate their precise location and visual information to a central monitoring system in real-time.

Crack Detection: The system employs the principle of ultrasonic sensing for non-contact crack detection. An ultrasonic sensor emits high-frequency sound waves and measures the time it takes for the echo to return. A significant reduction in the reflected signal strength or an unexpected reflection pattern indicates a potential crack or discontinuity on the track surface. This method is chosen for its ability to detect surface and near-surface flaws without physical contact, allowing for continuous monitoring during robot movement.

Obstacle Detection: Similar to crack detection, ultrasonic sensors are utilized for obstacle detection. By continuously measuring the distance to objects in front of the robot, the system can identify obstructions on the track. A sudden decrease in the measured distance below a predefined threshold triggers an obstacle detection event. This proactive approach aims to prevent derailments and ensure the safe passage of trains.

Localization and Communication: To provide actionable information, the detected anomalies need to be accurately located and reported. A Global Positioning System (GPS) module is integrated to determine the precise geographical coordinates of the robot when a crack or obstacle is detected. This location data, along with a timestamp, is crucial for maintenance teams to quickly identify and address the issue. A GSM (Global System for Mobile Communications) module facilitates real-time communication of this location information to a central server or designated personnel via SMS or GPRS. This ensures timely alerts and reduces response times.

Visual Verification: To provide richer contextual information about the detected anomaly, an ESP32-CAM module is incorporated. Upon detecting a potential crack or obstacle, the robot captures a high-resolution image of the affected area allows maintenance personnel to visually assess the severity and nature of the problem before physical inspection, optimizing resource allocation and repair strategies.

System Integration and Autonomy: The Node MCU (ESP8266) acts as the central microcontroller, orchestrating the operation of all the sensors and communication modules. It processes the sensor data, triggers the ESP32-CAM when an anomaly is suspected, interfaces with the GPS and GSM modules for location tracking and data transmission, and controls the robot's movement (though the locomotion mechanism is not the primary focus of this theoretical concept).

3.2 Block Diagram

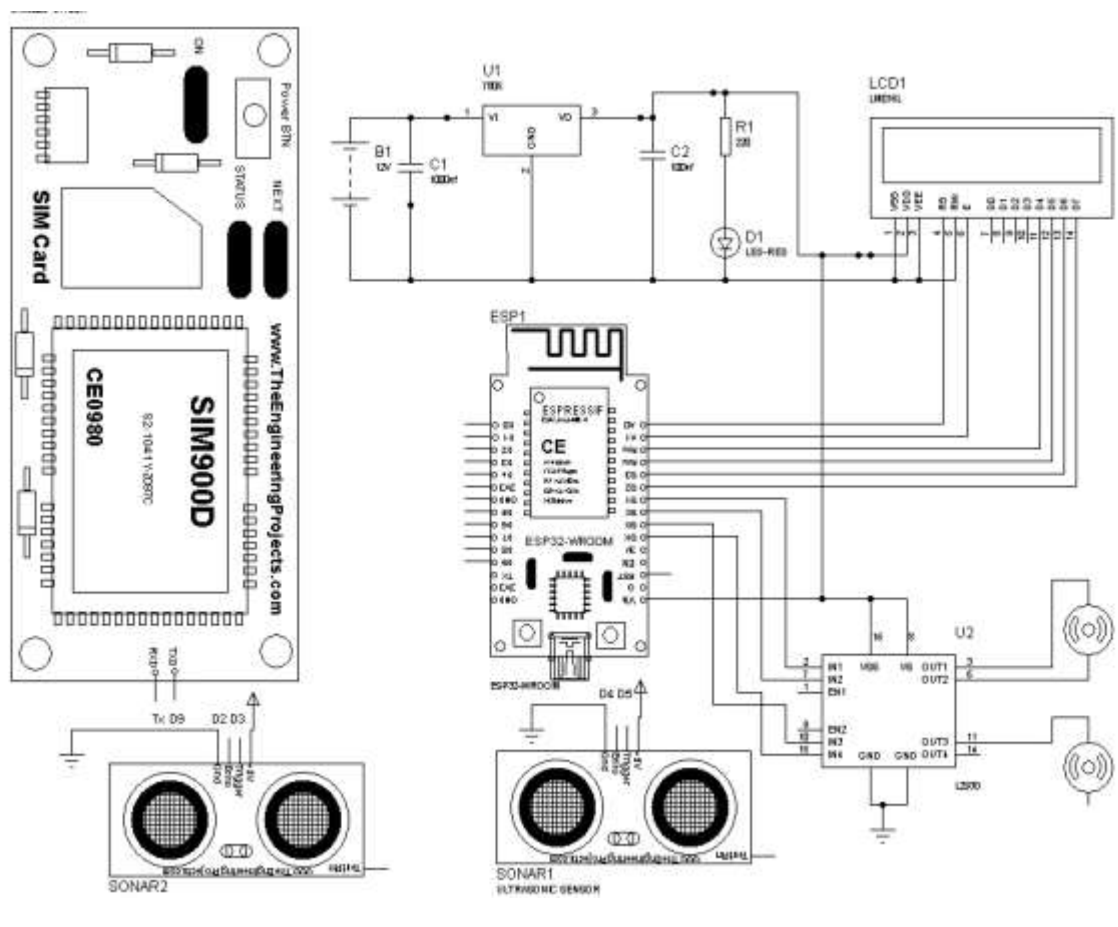

Fig 3.1 Block diagram of the railway detection system

3.3 Algorithm/Flowchart

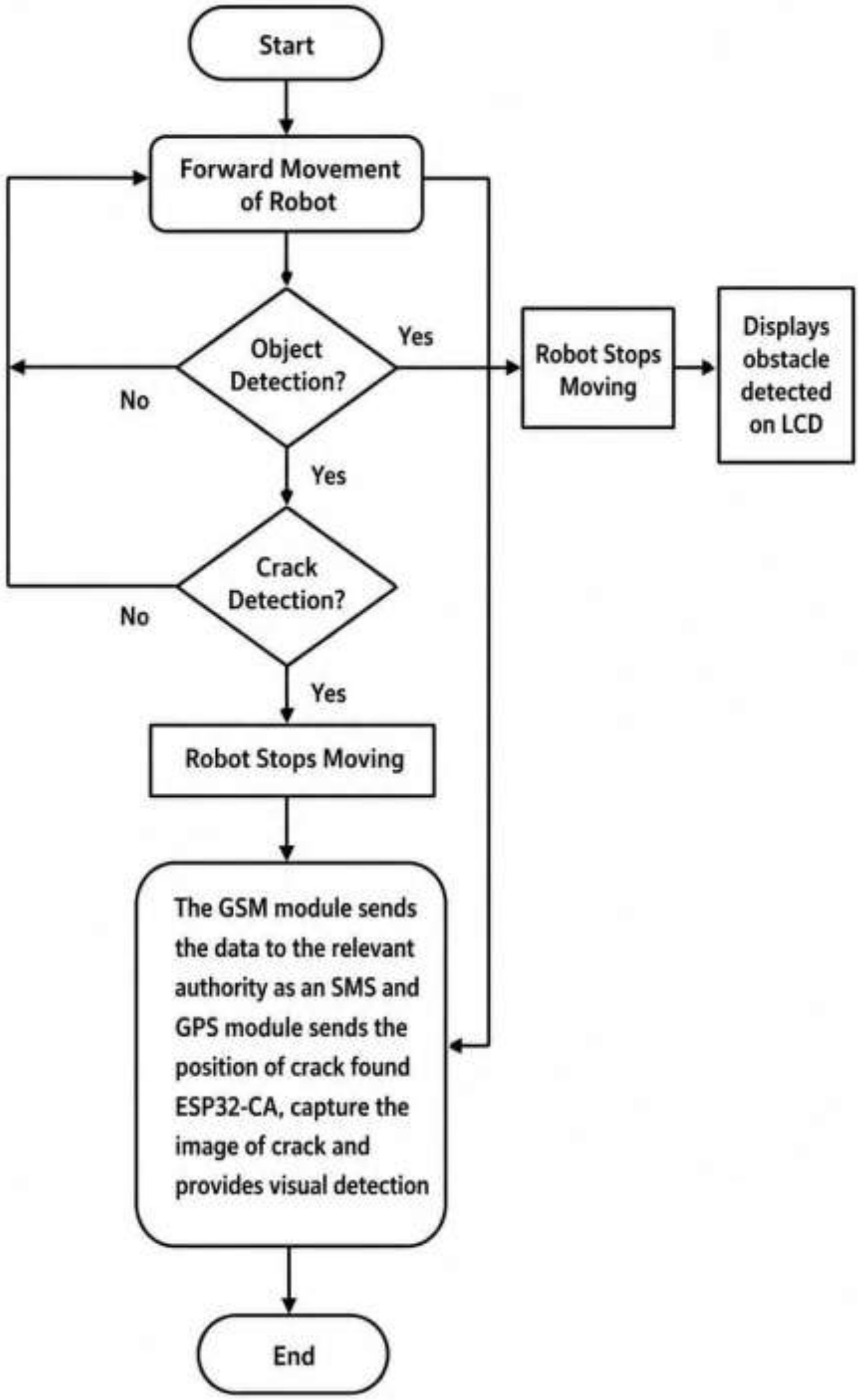

Fig 3.2: Flowchart of the railway defect detection system

3.3 Summary:

Theoretical Aspects of the Project," establishes the foundational concepts and system design for the automated railway track crack and object detection system. It outlines the theoretical underpinnings of image processing techniques for identifying anomalies from ESP32-CAM images and the integration of the ESP32-CAM with the NodeMCU for processing and control. The chapter details the system's architecture through a block diagram, illustrating the interconnectedness of the ESP32-CAM, NodeMCU, GPS module for location tagging, and GSM module for remote alert transmission. Furthermore, it presents the operational logic via an algorithm or flowchart, describing the sequential steps from image capture and analysis to location acquisition and alert dissemination upon detection of a crack or obstruction.

CHAPTER -IV: METHODOLOGY

4.1 Components Required:

4.1.1 LCD:

A liquid crystal display (LCD) is a thin, flat display device made up of any number of colour or monochrome pixels arrayed in front of a light source or reflector. Each pixel consists of a column of liquid crystal molecules suspended between two transparent electrodes, and two polarizing filters, the axes of polarity of which are perpendicular to each other. Without the liquid crystals between them, light passing through one would be blocked by the other. The liquid crystal twists the polarization of light entering one filter to allow it to pass through the other.

(1) Interface with either 4-bit or 8-bit microprocessor.

(2) Display data RAM

(3) 80x8 bits (80 characters).

(4) Character generator ROM

(5) 160 different 5 *7 dot-matrix character patterns.

(6) Character generator RAM

(7) 8 different user programmed 5 *7 dot-matrix patterns.

(8) Display data RAM and character generator RAM may be Accessed by the microprocessor.

(9) Numerous instructions

(10) Clear Display, Cursor Home, Display ON/OFF, Cursor ON/OFF, Blink Character, Cursor Shift, Display Shift.

(11). Built-in reset circuit is triggered at power ON.

(12). Built-in oscillator.

Fig 4.1: 16*2 LCD display

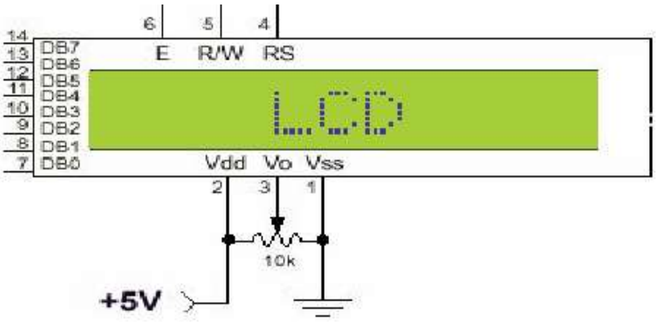

Fig 4.2 : pin diagram of 1x16 lines lcd

Most LCDs with 1 controller has 14 Pins and LCDs with 2 controller has 16 Pins (two pins are extra in both for back-light LED connections).

4.1.1.1 Pin Configuration of LCD:

PIN	SYMBOL	FUNCTION
1	Vss	Power Supply(GND)
2	Vdd	Power Supply(+5V)
3	Vo	Contrast Adjust
4	RS	Instruction/Data Register Select
5	R/W	Data Bus Line
6	E	Enable Signal
7-14	DB0-DB7	Data Bus Line
15	A	Power Supply for LED B/L(+)
16	K	Power Supply for LED B/L(-)

Table 4.1: Pins symbols and its functions.

CONTROL LINES:

EN: Line is called "Enable." This control line is used to tell the LCD that you are sending it data. To send data to the LCD, your program should make sure this line is low (0) and then set the other two control lines and/or put data on the data bus.When the other lines are completely ready, bring EN high (1) and wait for the minimum amount of time required by the LCD datasheet (this varies from LCD to LCD), and end by bringing it low (0) again.

RS: Line is the "Register Select" line. When RS is low (0), the data is to be treated as a command or special instruction (such as clear screen, position cursor, etc.). When RS is high (1), the data being sent is text data which sould be displayed on the screen. For example, to display the letter "T" on the screen you would set RS high.

RW: Line is the "Read/Write" control line. When RW is low (0), the information on the data bus is being written to the LCD. When RW is high (1), the program is effectively querying (or reading) the LCD. Only one instruction ("Get LCD status") is a read command. All others are write commands, so RW will almost always be low.Finally, the data bus consists of 4 or 8 lines (depending on the mode of operation selected by the user). In the case of an 8-bit data bus, the lines are referred to as DB0, DB1, DB2, DB3, DB4, DB5, DB6, and DB7.

Logic status on control lines:

- E - 0 Access to LCD disabled - 1 Access to LCD enabled.
- R/W - 0 Writing data to LCD - 1 Reading data from LCD.
- RS - 0 Instructions

4.1.2 BATTERY

lithium-ion battery or Li-ion battery (abbreviated as LIB) is a type of rechargeable battery in which lithium ions move from the negative electrode to the positive electrode during discharge and back when charging. Li-ion batteries use an intercalated lithium compound as one electrode material, compared to the metallic lithium used in a non-rechargeable lithium battery. The electrolyte, which allows for ionic movement, and the two electrodes are the constituent components of a lithium-ion battery cell.

Lithium-ion batteries are common in-home electronics. They are one of the most popular types of rechargeable batteries for portable electronics, with a high energy density, tiny memory effect[9] and low self-discharge. LIBs are also growing in popularity for military, battery electric vehicle and aerospace applications.

For example, lithium-ion batteries are becoming a common replacement for the lead–acid batteries that have been used historically for golf carts and utility vehicles. Instead of heavy lead plates and acid electrolyte, the trend is to use lightweight lithium-ion battery packs that can provide the same voltage as lead-acid batteries, so no modification to the vehicle's drive system is required.

Chemistry, performance, cost and safety characteristics vary across LIB types. Handheld electronics mostly use LIBs based on lithium cobalt oxide (LiCoO which offers high energy density, but presents safety risks, especially when damaged. Lithium iron phosphate (LiFePO lithium ion manganese oxide battery (Li Mn 2O) offer lower energy density, but longer lives and inherent safety. Such batteries are widely used for electric tools, medical equipment and other roles. NMC in particular is a leading contender for automotive applications. Lithium nickel cobalt aluminum oxide (LiNiCoAlO).Lithium-ion batteries can pose unique safety hazards since they contain a flammable electrolyte and may be kept pressurized.

4.1.3 GSM MODULE

Whether you want to monitor your home from afar or activate the sprinkler system in your garden with a missed call; then the SIM800L GSM/GPRS module can serve as a solid launching point. The SIM800L GSM/GPRS module is a miniature GSM modem that can be used in a variety of IoT projects.

You can use this module to do almost anything a normal cell phone can do, such as sending SMS messages, making phone calls, connecting to the Internet via GPRS, and much more.To top it all off, the module supports quad-band GSM/GPRS networks, which means it will work almost anywhere in the world.

At the heart of the module is a SIM800L GSM cellular chip from Simcom.The operating voltage of the chip ranges from 3.4V to 4.4V, making it an ideal candidate for direct LiPo battery supply. This makes it an excellent choice for embedding in projects with limited space.

Fig 4.3: GSM module

Fig 4.4 GSM labelling

All the necessary data pins of the SIM800L GSM chip are broken out to a 0.1″ pitch headers, including the pins required for communication with the microcontroller over the UART. The module supports baud rates ranging from 1200 bps to 115200 bps and features automatic baud rate detection.

The module requires an external antenna in order to connect to the network. So the module usually comes with a helical antenna that can be soldered to it. The board also has a U.FL connector If you wish to keep the antenna at a distance from the board.

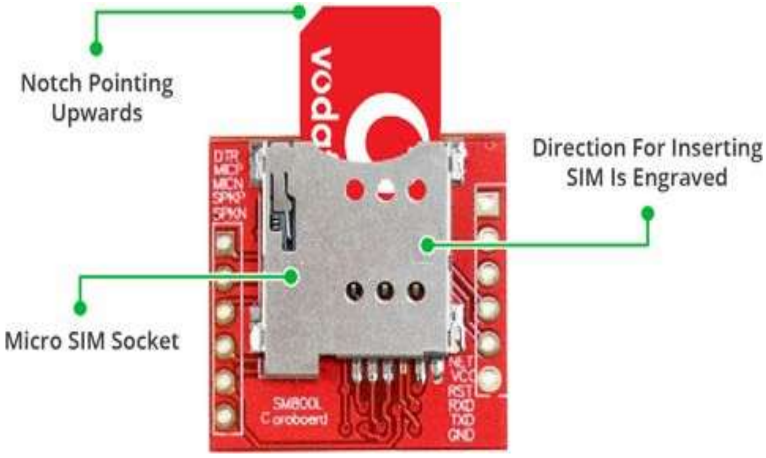

Fig 4.5 GSM module labelling

There's a SIM socket on the back! Any 2G Micro SIM card will work perfectly. The proper way to insert the SIM card is typically engraved on the surface of the SIM socket.

LED Status Indicators:

The SIM800L module has an LED that indicates the status of your cellular network. It will blink at different rates depending on the state it is in.

Blink every 1s: The chip is running but hasn't made a connection to the cellular network yet.

Blink every 2s: The GPRS data connection you reque sted is active.

Blink every 3s: The module has made contact with the cellular network and can send/receive voice and SMS.

Fig 4.6: Closed view of SIM 800

Fig 4.7:SIM800L

Choosing an Antenna:

The SIM800L module requires an external antenna in order to connect to the network, so choosing the right antenna is very important. There are two options available. The first is a helical antenna that comes with the module and can be soldered directly to the PCB. This antenna is very useful for space-constrained projects. However, be aware that you may face difficulties establishing a connection, particularly if your project is indoors.

Fig 4.8:GSM Module

Fig 4.9: SIM800L with antenna

Fig 4.10: GSM Module Pin out

Another option is a 3dBi GSM antenna with DRa U.FL to SMA adapter, which can be found online for less than $3. You can snap-fit this antenna into the small u.fl connector located on the top-left corner of the module. This type of antenna provides better performance and even allows your module to be placed inside a metal box as long as the antenna is outside.

Features:

- Supports Quad-band: GSM850, EGSM900, DCS1800 and PCS1900
- Connect onto any global GSM network with any 2G SIM
- Make and receive voice calls using an external 8Ω speaker & electret microphone
- Send and receive SMS messages
- Send and receive GPRS data (TCP/IP, HTTP, etc.)
- Scan and receive FM radio broadcasts
- Transmit Power:
- Serial-based AT Command Set
- FL connectors for cell antennae

4.1.4 ULTRASONIC SENSOR:

Ultrasonic Sensor HC-SR04 with Arduino:

This article is a guide about the popular Ultrasonic Sensor HC – SR04. The HC-SR04 ultrasonic sensor uses sonar to determine distance to an object like bats do. It offers excellent non-contact range detection with high accuracy and stable readings in an easy-to-use package. It comes complete with ultrasonic transmitter and receiver modules.Ultrasonic sensors use sound waves, beyond the range of human hearing, to measure distance and detect objects without physical contact. They emit a sound pulse and measure the time it takes for the echo to return, allowing them to calculate the distance to an object. These sensors are used in various applications like proximity sensing, level detection, and even in medical imaging.

Fig 4.11: Ultra sonic sensor

WORKING:

The transmitter (trig pin) sends a signal: a high-frequency sound. When the signal finds an object, it is reflected and the transmitter (echo pin) receives it. The time between the transmission and reception of the signal allows us to calculate the distance to an object. This is possible because we know the sound's velocity in the air.

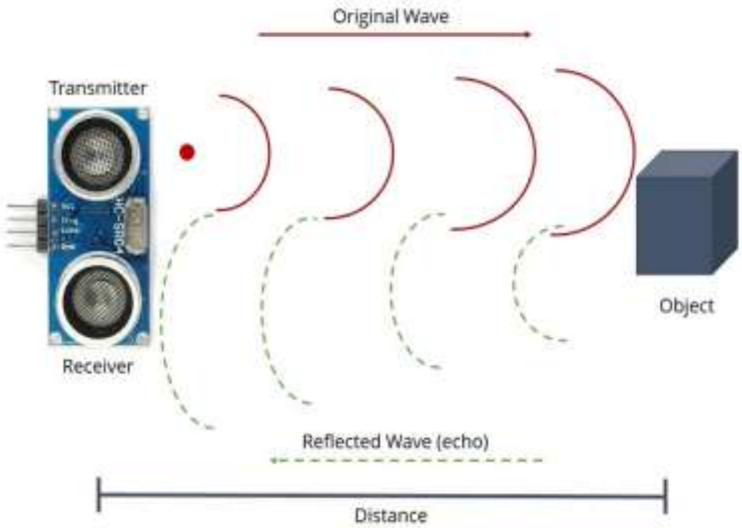

Fig 4.12: Working of UV

Pins:

- VCC: +5VDC
- Trig : Trigger (INPUT)
- Echo: Echo (OUTPUT)
- GND: GND

Features:

- Here’s a list of some of the HC-SR04 ultrasonic sensor features and specs:
- Power Supply :+5V DC
- Quiescent Current : <2mA
- Working Current: 15mA
- Effectual Angle: <15°
- Ranging Distance : 2cm – 400 cm/1″ – 13ft
- Resolution : 0.3 cm
- Measuring Angle: 30 degree
- Trigger Input Pulse width: 10Us
- Dimension: 45mm x 20mm x 15mm

4.1.5 L293D (QUADRUPLE HALF H-DRIVERS):

The L293 and L293D are quadruple high-current half-H drivers. The L293 is designed to provide bidirectional drive currents of up to 1 A at voltages from 4.5 V to 36 V. The L293D is designed to provide bidirectional drive currents of up to 600-mA at voltages from 4.5 V to 36 V. All inputs are TTL compatible.

Features:

- Wide Supply-Voltage Range: 4.5 V to 36 V
- Separate Input-Logic Supply
- Internal ESD Protection
- Thermal Shutdown
- High-Noise-Immunity Inputs

4.1.6 DC MOTOR:

A DC motor is designed to run on DC electric power. Two examples of pure DC designs are Michael Faraday's homopolar motor (which is uncommon), and the ball bearin motor, which is (so far) a novelty. By far the most common DC motor types are the brushed and brushless types, which use internal and external commutation respectively to create an oscillating AC current from the DC source -- so they are not purely DC machines in a strict sense.

Types of dc motors:

1. Brushed DC Motors
2. Brushless DC motors
3. Coreless DC motors

Brushed DC motors:

The classic DC motor design generates an oscillating current in a wound rotor with a split ring commutator, and either a wound or permanent magnet statorSome of the problems of the brushed DC motor are eliminated in the brushless design. In this motor, the mechanical "rotating switch" or commutator/brushgear assembly is replaced by an external electronic switch synchronised to the rotor's position. Brushless motors are typically 85-90% efficient, whereas DC motors with brushgear are typically 75-80% efficient. Midway between ordinary DC motors and stepper motors lies the realm of the brushless DC motor.

Coreless DC motors:

Nothing in the design of any of the motors described above requires that the iron (steel) portions of the rotor actually rotate; torque is exerted only on the windings of the electromagnets. Taking advantage of this fact is the coreless DC motor, a specialized form of a brush or brushless DC motor. Optimized for rapid acceleration, these motors have a rotor that is constructed without any iron core. The rotor can take the form of a winding-filled cylinder inside the stator magnets, a basket surrounding the stator magnets, or a flat *pancake* (possibly formed on a printed wiring board) running between upper and lower stator magnets. Filled epoxies that have moderate mixed viscosity and a long gel time. These systems are highlighted by low shrinkage and low exotherm. Typically UL 1446 recognized as a potting compound for use up to 180C (Class H) UL File No. E 210549.

4.1.7 ESP32-CAM:

This article is a quick getting started guide for the ESP32-CAM board. We'll show you how to setup a video streaming web server with face recognition and detection in less than 5 minutes with Arduino IDE. The ESP32-CAM is a very small camera module with the ESP32-S Besides the OV2640 camera, and several GPIOs to connect peripherals, it also features a microSD card slot that can be useful to store images taken with the camera or to store files to serve to clients.

Fig 4.13: Esp32

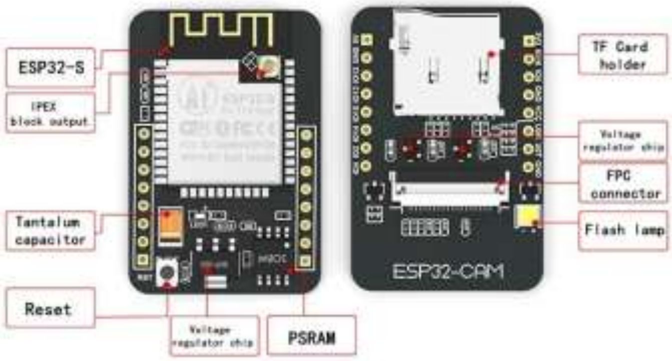

Fig 4.14: Esp32 labling

The smallest 802.11b/g/n Wi-Fi BT SoC module

- Low power 32-bit CPU, can also serve the application processor
- Up to 160MHz clock speed, summary computing power up to 600 DMIPS
- Built-in 520 KB SRAM, external 4MPSRAM
- Supports UART/SPI/I2C/PWM/ADC/DAC
- Support OV2640 and OV7670 cameras, built-in flash lamp
- Support image Wi FI upload
- Support TF card
- Supports multiple sleep modes
- Embedded Lwip and Free RTOS
- Supports STA/AP/STA+AP operation mode

There are three GND pins and two pins for power: either 3.3V or 5V.GPIO 1 and GPIO 3 are the serial pins. You need these pins to upload code to your board. Additionally, GPIO 0 also plays an important role, since it determines whether the ESP32 is in flashing mode or not. When GPIO 0 is connected to GND, the ESP32 is in flashing mode.

The following pins are internally connected to the microSD card reader:

- GPIO 14: CLK
- GPIO 15: CMD
- GPIO 2: Data 0
- GPIO 4: Data 1 (also connected to the on-board LED)
- GPIO 12: Data 2
- GPIO 13: Data 3

4.1.8 LM2596:

This tutorial will show how to use LM2596 Buck Converter to power up devices requiring different voltages. We will show which are the best types of batteries to use with the converter and how to get more than just one output from the converter (indirectly).

This is our first tutorial in our series on Power Distribution, we believe that Power Distribution is often overlooked and that this is a big reason why many people lose interest in robotics in the beginning, for instance they burn up their components and are unwilling to buy new components from the fear to just burn them up again, we hope that this series on Power Distribution will help you understand how to better work with electricity.

Fig 4.15: LM2596

4.1.9 GPS

The Global Positioning System (GPS) is a satellite-based navigation system made up of at least 24 satellites. GPS works in any weather conditions, anywhere in the world, 24 hours a day, with no subscription fees or setup charges.

Fig 4.16: GPS

How GPS works:

GPS satellites circle the Earth twice a day in a precise orbit. Each satellite transmits a unique signal and orbital parameters that allow GPS devices to decode and compute the precise location of the satellite. GPS receivers use this information and trilateration to calculate a user's exact location. Essentially, the GPS receiver measures the distance to each satellite by the amount of time it takes to receive a transmitted signal.

NEO-6M GPS Chip:

The heart of the module is a NEO-6M GPS chip from u-box. It can track up to 22 satellites on 50 channels and achieves the industry's highest level of sensitivity i.e. -161 dB tracking, while consuming only 45mA supply current. The u-box 6 positioning engine also boasts a Time-To-First-Fix (TTFF) of under 1 second. One of the best features the chip provides is Power Save Mode (PSM). It allows a reduction in system power consumption by selectively switching parts of the receiver ON and OFF.

Fig 4.17: NEO-6M GPS Chip

Position Fix LED Indicator: There is an LED on the NEO-6M GPS Module which indicates the status of Position Fix. It'll blink at various rates depending on what state it's in No Blinking ==> means It is searching for satellites Blink every 1s – means Position Fix is found.

3.3V LDO Regulator

The operating voltage of the NEO-6M chip is from 2.7 to 3.6V. But, the module comes with MIC5205 ultra-low dropout 3V3 regulator from MICREL. The logic pins are also 5-volt tolerant, so we can easily connect it to an Arduino or any 5V logic microcontroller without using any logic level converter.

4.1.10 Node MCU

Node MCU is an open source IoT platform. It includes both firmware which runs on the ESP8266 Wi-Fi SoC, and hardware which is based on the ESP-12 module.The applications in these samples that are running on Node MCU are written using Lua scripting language which is quite simple and easy to understand.

Node MCU is an open-source firmware and development kit that helps you to prototype or build IoT products. It includes firmware that runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which is based on the ESP-12 module. The firmware uses the Lua scripting language. It is based on the eLua project and built on the Espressif Non-OS SDK for ESP8266.

Fig 4.18: MCU

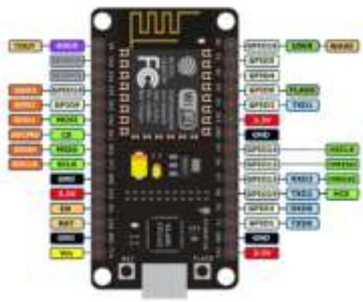

Fig 4.19: Node MCU labelling

MCU stands for Micro Controller Unit - which really means it is a computer on a single chip. A microcontroller contains one or more CPUs (processor cores) along with memory and programmable input/output peripherals. They are used to automate automobile engine control, implantable medical devices, remote controls, office machines, appliances, power tools, toys etc.

It includes firmware that runs on the ESP8266 Wi-Fi SoC from Espressif Systems, and hardware which is based on the ESP-12 module. The firmware uses the Lua scripting language. It is based on the eLua project and built on the Espressif Non-OS SDK for ESP8266. Node MCU is an open source IoT platform.

NodeMCU is an open-source IoT platform featuring both firmware and hardware based on the ESP8266 Wi-Fi SoC. It's designed for prototyping and building IoT devices, using the Lua scripting language and compatible with the Arduino IDE.

4.2 Implementation:

Here the proposed model is made up of hardware which was previously explained in the description of the system designed hardware.

Fig 4.20: Proposed Method

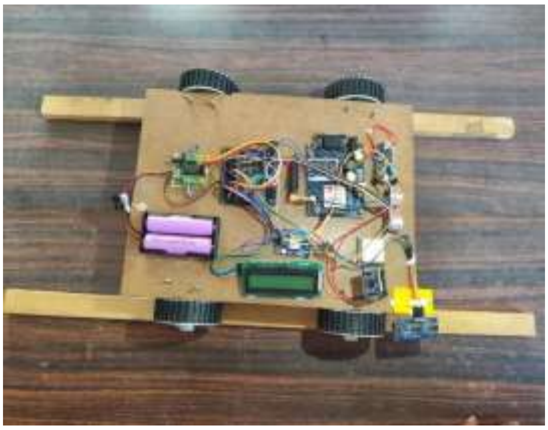

Fig 4.21: Proposed method with wooden track

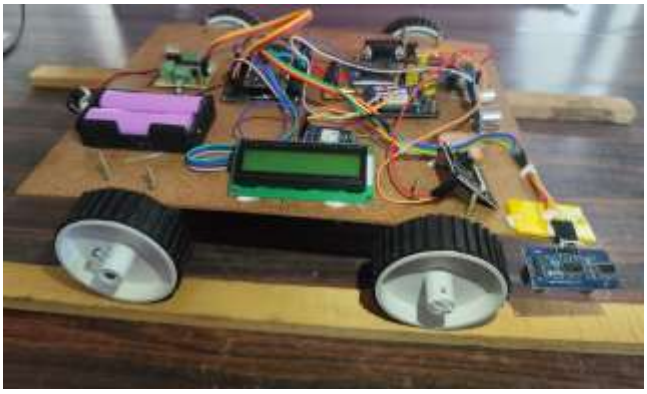

Fig 4.22: Side view of proposed method

4.3 Summary:

"Methodology," details the practical implementation of the railway track crack and object detection system. It begins by listing the components required, providing a bill of materials that includes the LCD for potential local display, the battery for power, the GSM module for wireless communication, the ultrasonic sensor (likely for obstacle detection or distance measurement), the L293D motor driver and DC motor (suggesting a mobile platform), the ESP32-CAM for image acquisition, the IM2596 voltage regulator for power management, the GPS module for location tracking, and the NodeMCU as the central microcontroller. Following this, the chapter outlines the implementation process, describing the steps involved in assembling the hardware components, integrating the software for image processing, GPS and GSM communication, and potentially detailing the development of the mobile platform's control. This section would likely cover the physical connections, the programming environment, and the overall construction of the prototype system. In essence, Chapter 4 bridges the theoretical aspects with the tangible realization of the project.

CHAPTER-V

RESULTS

5.1 Results:

The following figure shows that the SMS obtained on the mobile phone with the latitudinal and longitudinal position at the point where a crack or obstacle is detected.

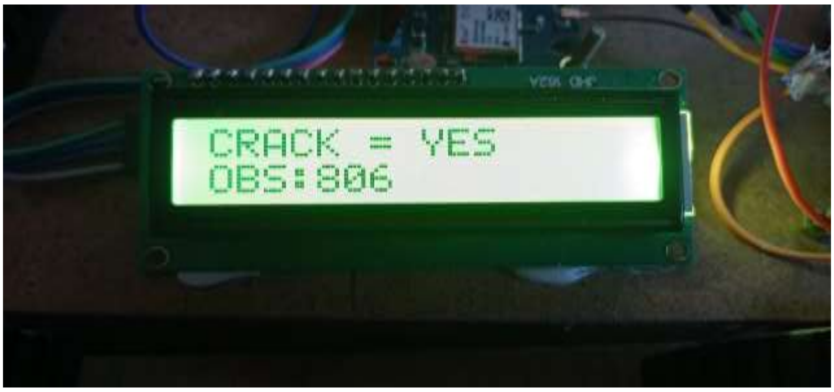

Fig 5.1: Crack detected

Fig 5.2: Obstacle detected


RIGHT TRACK CRACK DETECTED AT https://www.google.com/maps/place/17.459755,78.557289

Fig 5.3: SMS received in the mobile phone

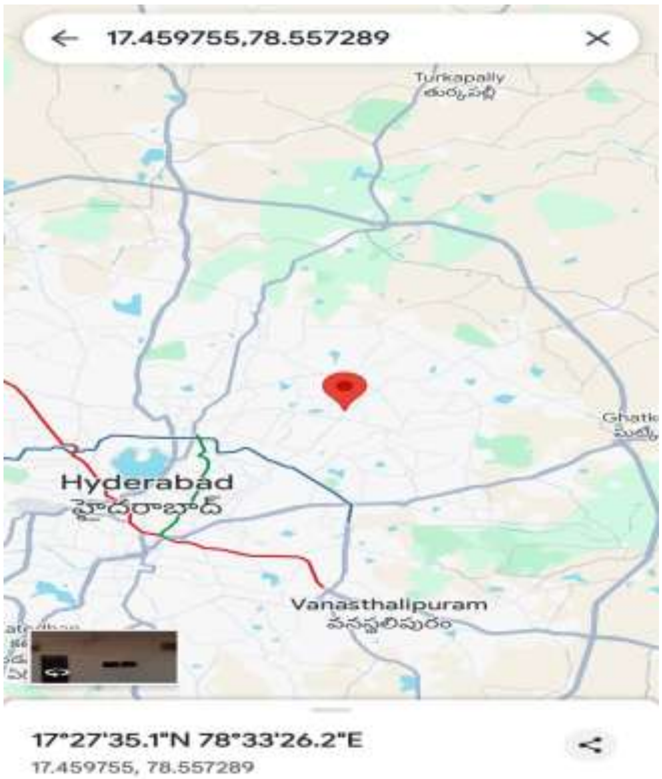

Fig 5.4: Location of the crack detected in the google map

Fig 5.5: Crack visual inspection in Esp32 cam

CHAPTER –VI

SUMMARY AND CONCLUSION

6.1 Conclusion:

This project successfully demonstrated a functional prototype for railway track crack and obstacle detection. By integrating ultrasonic sensors for real-time detection, GPS and GSM modules for immediate location reporting, and the ESP32 for image capture and transmission, the system offers a comprehensive approach to enhancing railway safety. The collected data, including precise location and visual confirmation, empowers timely maintenance and proactive intervention, ultimately contributing to a safer and more efficient railway network. Further development could focus on refining sensor accuracy, optimizing power consumption, and exploring integration with existing railway management systems for seamless operation.

6.2 Future Scope:

The future of railway track crack and object detection systems leveraging GPS for precise location tracking and GSM for instant alerts is poised for significant advancement through the integration of sophisticated sensors like LiDAR, high-resolution cameras with AI-powered image processing, and acoustic/vibration sensors. These systems will utilize increasingly powerful and energy-efficient microcontrollers, potentially incorporating edge AI capabilities, to analyze sensor data for improved detection accuracy and reduced false positives. Integration with centralized railway management systems will enable automated work order generation and predictive maintenance, while potential fusion with Positive Train Control and level crossing safety systems will further enhance overall safety. Autonomous inspection vehicles equipped with this technology promise more frequent and comprehensive track monitoring, leading to safer, more efficient, and more reliable railway transportation.

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APPENDIX

```
#include <SoftwareSerial.h>
#include <ESP8266WiFi.h>
#include <stdlib.h>
#include <LiquidCrystal_I2C.h>
#include <OneWire.h>
#include <Wire.h>
#include <TinyGPS++.h>
TinyGPSPlus gps;
LiquidCrystal_I2C lcd(0x27, 16, 2);
const int m11 = D0;
const int m12 = D3;
const int bz = D4;
float lat = 17.459755;
float logt = 78.557290;//home
String mobilenumber1 = "AT+CMGS=\"+917396093171\"\r";
const int trigPine_track= D5;
const int echoPine_track= D6;
const int trigPine_obs= D7;
const int echoPine_obs= D8;
long duration, distance,cm_in,cm_in_obs,inches_in;
void setup()
{
  Serial.begin(9600);
  Wire.begin(D2,D1);//sda,scl
  pinMode(m11,OUTPUT);
  pinMode(m12,OUTPUT);
  pinMode(bz,OUTPUT);
  pinMode(trigPine_track, OUTPUT);
  pinMode(echoPine_track, INPUT);
  pinMode(trigPine_obs, OUTPUT);
```

```
  pinMode(echoPine_obs, INPUT);
  digitalWrite(m11,LOW);
  digitalWrite(m12,LOW);
  digitalWrite(bz,LOW);
  lcd.begin();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("RAILWAY TRACK ");
  lcd.setCursor(0, 1);
  lcd.print("CRACK DETECTION..");
  delay(2000);
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("TRACK:");
  lcd.setCursor(0, 1);
  lcd.print("OBS: ");
  Serial.println("AT");
  Serial.println("AT");
  delay(1000);
  Serial.println("AT+CNMI=2,2,0,0,0");
  delay(500);
  Serial.println("AT+CMGF=1");
  get_gps();
  delay(1000);
  lcd.clear();
}
void loop()
{
  UltrasonicSensor(trigPine_track, echoPine_track);
  cm_in = distance;
  Serial.println(distance);
  lcd.setCursor(0, 0);
  lcd.print("TRACK:");
```

```
lcd.print(distance);
lcd.print("       ");
UltrasonicSensor(trigPine_obs, echoPine_obs);
cm_in_obs = distance;
Serial.println(cm_in_obs);
lcd.setCursor(0, 1);
lcd.print("OBS:");
lcd.print(cm_in_obs);
lcd.print("      ");
if(cm_in > 4)
{
stop();
lcd.setCursor(0,0);
lcd.print("CRACK = YES    ");
stop();
digitalWrite(bz,HIGH);
send_sms(mobilenumber1,"RIGHT TRACK CRACK DETECTED AT");
//Serial.println("left YES");
}
else
{
if(cm_in_obs > 20)
{
start();
}
else
{
lcd.setCursor(0,1);
lcd.print("OBJECT = YES");
stop();
}
}
delay(500);
```

```
}
void start()
{
  digitalWrite(m11,HIGH);
  digitalWrite(m12,HIGH);
}
void stop()
{
  digitalWrite(m11,LOW);
  digitalWrite(m12,LOW);
}
void send_sms(String mnumber,String str)
{
   Serial.println("AT");
   delay(500);
   Serial.println("AT+CMGF=1");
   delay(500);
   //gsm.println("AT+CMGS=\"+916300102942\"\r");
   Serial.println(mnumber);
   delay(1000);
   //Serial.println("TROUBLE                                                    AT
https://www.google.com/maps/place/15.4340,80.0467");
   Serial.print(str);
   Serial.print(" https://www.google.com/maps/place/");
   Serial.print(lat,6);
   Serial.print(",");
   Serial.println(logt,6);
   delay(2000);
   Serial.println((char)26); // ASCII code of CTRL+Z
   delay(1000);
   Serial.println("AT");
}
void get_gps()
```

```
{
 lcd.clear();
 for(int i=0;i<80;i++)
 {
while (Serial.available() > 0){
   gps.encode(Serial.read());
   if(gps.location.isUpdated()){
    /* Serial.print("Latitude= ");
     Serial.print(gps.location.lat(), 6);
     Serial.print(" Longitude= ");
     Serial.println(gps.location.lng(), 6);*/
     lcd.setCursor(0,0);
     lcd.print("LAT:");
     lcd.print(gps.location.lat(), 6);
     lcd.setCursor(0,1);
     lcd.print("LOG:");
     lcd.print(gps.location.lng(), 6);
     lat = gps.location.lat();
     logt = gps.location.lng();
     delay(1000);
   }
 }
 delay(80);
}
     lcd.setCursor(0,0);
     lcd.print("LAT:");
     lcd.print(lat, 6);
     lcd.setCursor(0,1);
     lcd.print("LOG:");
     lcd.print(logt, 6);
}
void UltrasonicSensor(int trigPin, int echoPin)
{
```

```
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = (duration / 2) / 29.1;
}
```